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TECHNICAL SUPPLEMENT TO NEURAL BELLMAN OPERATORS

QIAN QI
Peking University

This supplement retains optimal-advantage centering and every preceding complete proof. It adds the qualification-readout primal and dual, the original investment cover and actor calculations, and the predictable original-law cost and optional-stopping proofs. Detailed scientific records and historical economic sections remain in the Numerical and Development Appendix.

1. POLICY-COST INFERENCE AT THE SCALE OF VERIFIED REGRET

The query certificate and a direct cost experiment answer different questions. Nevertheless, a certificate can improve the experiment without replacing its estimand. The relevant observation is a sum of optimal Bellman advantages along the actual policy trajectory. Its expectation is policy loss, while its range is controlled by the local accuracy contract rather than by the range of economic costs. All conditional expectations used to construct this observation must themselves be evaluated and charged.

Write V_t for the original Bellman optimum, $V_T = g$, $b_0 = 1$, and $b_t = \prod_{j<t} \beta_j$. For a feasible frozen policy π, define

$$a_t^\pi(x) = c_t(x, \pi_t(x)) + \beta_t P_t^{\pi_t(x)} V_{t+1}(x) - V_t(x), \qquad Z^\pi = \sum_{t<T} b_t a_t^\pi(X_t^\pi).$$

The symbol a_t^π denotes an advantage, not an action. Assume a verified local bound $0 \le a_t^\pi(x) \le e_t^\pi$ for every true state, including the acquisition allowance. Put $G_\pi = \sum_{t<T} b_t e_t^\pi$. A sample contains the actual trajectory of each compared controller under the original law; common initial states and common innovations may couple the controllers. Different sample rows are independent conditional on the frozen construction.

Qian QI: qiqian@pku.edu.cn

THEOREM 1—Bellman-centered cost comparison: *For two such controllers π, ν with a common initial law λ, let $D = Z^\pi - Z^\nu$. Then*

$$\mathbb{E}D = \mathbb{E}_\lambda[J_0^\pi - J_0^\nu], \qquad -G_\nu \le D \le G_\pi. \tag{1}$$

Suppose deterministic computation encloses D_i by $[L_i, U_i] \subseteq [-G_\nu, G_\pi]$ on every sampled continuous bin, including every compatible acquired action. For a conditionally fixed finite family of M contrasts, put $R = G_\pi + G_\nu$ and

$$h_n = R\sqrt{\frac{\log(2M/\alpha)}{2n}}.$$

With probability at least $1 - \alpha$, simultaneously for the family,

$$\mathbb{E}_\lambda[J_0^\pi - J_0^\nu] \in \left[\frac{1}{n}\sum_i L_i - h_n, \frac{1}{n}\sum_i U_i + h_n\right]. \tag{2}$$

The interval may be intersected with $[-G_\nu, G_\pi]$. Its width is the mean numerical enclosure width plus $2h_n$. More generally, any valid endpoint-sample concentration bound may replace the bounded-sum radius.

Proof. See Supplement, Section 3.5.

The construction is related to martingale control variates and sequential value-based evaluation, including Jiang and Li (2016). It is not an off-policy importance-weighting argument and does not assume that a fitted value is correct. The additional requirements here are an original-optimum advantage enclosure, a true-state local regret bound, and a paid numerical evaluation of the conditional value. The target in (2) is the original expected cost difference, not the difference between two regret certificates. The realized centered observation generally differs from the realized cash-cost difference; their expectations agree by the displayed identity.

1.1. *Numerical construction and a target-precision rule*

The returned action is fixed before evaluation. At a center $\bar{x}$ of a true-state box, let $[Q_L, Q_U]$ enclose $Q_t^*(\bar{x}, \pi_t(x))$ for that fixed action and let $[V_L, V_U]$ enclose $V_t(\bar{x})$.

If L_Q, L_V are state Lipschitz bounds and ρ bounds $\|x - \bar{x}\|$, then

$$[Q_L - V_U - (L_Q + L_V)\rho, Q_U - V_L + (L_Q + L_V)\rho] \cap [0, e_t^\pi] \tag{3}$$

is a valid advantage enclosure. An empty intersection stops verification. Actions must be feasible at every state used by the endpoint queries. If the box crosses an acquisition boundary, one may enumerate all compatible acquired actions, or use $[0, e_t^\pi]$ for the unresolved part. The latter is conservative and its width remains in the reported numerical account. It is not permissible to discard that trajectory.

For the scalar investment economy, at r remaining dates we use

$$L_Q = \frac{8}{d} + \frac{11\beta}{16} G_{r-1}, \qquad L_V = G_r, \qquad e_t^\pi = \delta_t + d(L_Q + L_V)2^{-b},$$

from the original acquisition argument. The evaluator queries the optimum with its own, smaller tolerance; it does not retrain the policy or change the action being evaluated. Original-law quadrature, action-cover error and arithmetic are already included in $[Q_L, Q_U]$ and $[V_L, V_U]$. They must not be omitted or counted again as independent random noise.

COROLLARY 2—Precision and noninferiority: *Fix a numerical half-width allocation $v > 0$ and a sampling allocation $s > 0$ before observing the new contrasts. If $n^{-1}\sum_i (U_i - L_i) \le 2v$ and*

$$n \ge \frac{R^2 \log(2M/\alpha)}{2s^2},$$

then the interval in (2) has half-width at most $v + s$. For a prespecified loss margin $m > 0$, an upper endpoint at most m establishes noninferiority at that margin; containment in $[-m, m]$ establishes the corresponding two-sided equivalence statement on the same event. Neither statement identifies a strict advantage.

Proof. See Supplement, Section 3.6.

Both n and evaluator precision have costs. The deposited design fixes them prospectively and reports failure to meet the width allocation rather than increasing the sample after inspecting a favorable sign. The actual numerical width is

decomposed into the conditional Bellman enclosure, the state-bin Lipschitz expansion, unresolved acquisition branches and endpoint arithmetic. The conditional Bellman component includes its action-cover and continuous-shock quadrature work. A smaller statistical support is therefore not advertised as a free runtime improvement.

1.2. *The adoption decision after inference*

Let $[l, u]$ enclose $\mathbb{E}[J^\pi - J^\nu]$, let W_π, W_ν be measured complete deployment work under a declared boundary, and let $p \geq 0$ be a decision maker's price per unit of that work. Replacing ν by π at fee $f \geq 0$ has net-cost difference in

$$[l + p(W_\pi - W_\nu) + f,\ u + p(W_\pi - W_\nu) + f].$$

A negative upper endpoint licenses the replacement under those declared prices. A noninferiority finding permits a separate assessment of computational savings; it does not convert a normalized loss margin into an estimated monetary value. This keeps the original economic purchase question while making its statistical precision and its required price inputs explicit.

2. PROOFS OF THE INHERITED NBO RESULTS

These proofs retain the original statements and arguments. Their placement in the supplement changes the reading order, not their hypotheses.

2.1. *Proof of the result 1*

PROOF: Backward comparison and cash invariance give

$$V_t^* \geq f_t - \sum_{k=t}^{T-1} B_{t,k} u_k - B_{t,T} u_T, \quad J_t^\pi \leq f_t + \sum_{k=t}^{T-1} B_{t,k} v_k - B_{t,T} \ell_T,$$

where $B_{t,k} = \prod_{j=t}^{k-1} \beta_j$. The terminal claim is the stipulated enclosure; applying the Bellman or fixed-policy operator proves each inductive step. Subtraction gives (3). A shift $f_t \mapsto f_t + c_t^0$ adds $c_t^0 - \beta_t c_{t+1}^0$ to u_t and subtracts it from v_t. The terminal width is unchanged. *Q.E.D.*

2.2. *Proof of the result 2*

PROOF: The selected node label lies between the node Bellman infimum minus e_t and that infimum plus $D_t k_t + e_t$. The Lipschitz property of the infimum implies that every cone lies above it minus e_t. At a nearest node its cone exceeds the infimum by at most $D_t k_t + e_t + 2L_t h_t$. At the attaining original index, the two feasible pairs and (4) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \le y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t \zeta_t.$$

This proves (6). The terminal residual belongs to $[-e_T, e_T + 2L_g h_T]$; Proposition 1 gives the loss bound. A finite minimum of equally Lipschitz cones is equally Lipschitz. The identities $|z| = \mathrm{ReLU}(z) + \mathrm{ReLU}(-z)$ and $\min(u, v) = u - \mathrm{ReLU}(u - v)$ give the exact network. Finite lexicographic comparison and Borel repair give measurability. Effective refinement of the finitely many positively weighted terms yields the termination assertion. *Q.E.D.*

2.3. *Proof of the result 3*

PROOF: The triangle inequality bounds $f(x)$ above by each corner expression. Choose the lexicographically first attaining site $i_*(x)$. In every coordinate choose the cell endpoint between x_j and $x_{i_*,j}$. At this corner distances add exactly. An owner of strictly smaller transformed value would improve the objective at x; a tied smaller owner would contradict the original tie rule. Thus this corner has owner $i_*(x)$, proving both identities. Along a coordinate line, forward and backward relaxations respectively minimize over sites on either side, always comparing value–original-index pairs. Iterating over coordinates gives the separable distance transform and its stated work. Rational arithmetic and unchanged repair complete the claim. *Q.E.D.*

2.4. *Proof of the result 4*

PROOF: The state cost has gradient infinity norm at most $8/d$, Hessian norm at most $22/d$, and diagonal second derivatives at most $5/d + 16/d^2$. For the terminal cost the corresponding bounds are $10/d$, $26/d$, and $9/d + 16/d^2$. They follow by differentiating the quadratic terms and the continuously differentiable squared

shortage. The bound on the cycle Laplacian is four. The shortage Hessian, where it exists, is $16\mathbf{1}\mathbf{1}^\top/d^2$; the same second-difference bound holds across its kink.

For feasible (x, a) and (y, b), put $w = x - y$. The difference between the transition at their convex combination and the corresponding combination of transitions is $\theta(1-\theta)w_j w_{j+1}/16$ in coordinate j. Convexity and the G_{t+1} Lipschitz bound therefore give a convexity defect of at most

$$\frac{\beta G_{t+1}}{16}\theta(1-\theta)\sum_j |w_j w_{j+1}| \le \frac{\beta G_{t+1}}{16}\theta(1-\theta)\|w\|_2^2.$$

The state cost supplies curvature $4/d$, while the two-control action cost has Hessian eigenvalues at least $7/4$. Thus the state curvature of Q_t^* is at least $4/d - \beta G_{t+1}/8$, which is at least $107/(128d)$ when $G_{t+1} \le 27/d$. Partial minimization over the affine capacity graph preserves the state convexity inequality. The minimizing action is unique, including at a binding capacity constraint.

To propagate Lipschitz continuity, write $a = c(x)u$, where u lies in the fixed unit simplex. Transfer the same u from x to y. Each action-cost partial derivative has absolute value at most $13/16$, and $\sum_j u_j \le 1$. The action-cost transfer contributes at most $13\|x - y\|_1/(128d)$. The transition Jacobian at a fixed action has column sums at most $11/16$; the capacity transfer adds at most $3/64$. Comparing a minimizing normalized action in both directions gives the stated recursion for G_t. Its right side maps $27/d$ below itself.

For a fixed u, the transition as a function of x has Jacobian norm at most $11/16 + 1/16 = 3/4$. Its second-order composition term is bounded by $G_{t+1}/8$. The action-cost Hessian has norm at most $21/4$, and the capacity gradient has squared norm $1/(64d)$. These facts give the recursion for M_t. The infimum over u preserves this common semiconcavity bound: after subtracting the same quadratic, each fixed-u function is concave, and an infimum of concave functions is concave. The right side maps $54/d$ below itself. Convexity together with semiconcavity yields the claimed differentiability and gradient regularity; this conclusion does not require a differentiable optimizing-action map.

Along one state coordinate, the fixed-u transition has zero second derivative. The squared norm of its first derivative is at most $85/256 + 23/(256d)$: the two

original nonzero entries are bounded by $9/16$ and $1/8$, and the capacity-induced entries are bounded by $1/(16d)$. Combining this with the diagonal stage-cost bound gives D_t. Finally, positive interpolation along one coordinate has error between zero and $D_t h^2/8$. Applying the coordinate interpolations successively and adding the bounds proves (10). *Q.E.D.*

2.5. *Proof of the result 5*

PROOF: Joint convexity, including the feasible graph, gives

$$Q_t^*(x, \bar{\pi}_t(x)) \le \sum_v w_v(x) Q_t^*(v, a_t(v)) \le I_n q_t^{\mathrm{sel},+}(x).$$

The lower interpolation bound gives $V_t^*(x) \ge I_n l_t(x) - \kappa_t$. Subtraction proves the uniform optimal-advantage bound e_t. The partial action derivatives of Q_t^* are bounded by Λ_t, so feasible implementation error adds at most $\Lambda_t b_t$. Adding the future policy loss and proceeding backwards proves the recursion. No convexity of $J^{\widehat{\pi}}$ is used. *Q.E.D.*

2.6. *Proof of the result 6*

PROOF: Apply concavity to $f(a) - Ha^2/2$ and substitute the two lower endpoints. The displayed quadratic has derivative $D - K + 2Ks$. Minimizing it on the unit interval gives the three cases. Its coefficient of K is $-s(1-s)$, which proves the last assertion. *Q.E.D.*

2.7. *Proof of the result 7*

PROOF: At $r = 1$, analytic endpoint enclosures and the minorant lemma give the assertion. Suppose it holds for $r - 1$. Each child interval encloses the true optimal continuation. If the numerical next state is an interval, expanding the queried center value by G_{r-1} times the maximum ℓ^1 displacement encloses all its possible values. Conditional-mean lower bounds and the semiconcavity remainder then give valid endpoint brackets for the true Q_r. The minorants cover the entire action interval. A downward capacity endpoint can omit only an interval of length equal to the certified capacity-rounding difference; subtracting Λ_r times that difference accounts

for its possible improvement. The minimum upper endpoint is the cost of a feasible first action followed by the optimal future. This proves (i) by induction. Retaining earlier valid lower bounds preserves every conclusion.

For termination, an interval of length h has minorant at least its smaller lower endpoint minus $H_r h^2/8$. Its smaller upper endpoint is an available global upper bound. With the endpoint-width guard, an interval of length at most $h_\delta = \sqrt{3\delta/H_r}$ cannot be responsible for a gap exceeding δ. Every refined parent is therefore longer than h_δ. Its children are longer than $h_\delta/5$. Apart from the at most two initial fragments made by a prediction, disjoint terminal fragments consequently have total number at most $5c(x)/h_\delta$, up to endpoint counting. Equation (15) is a conservative point budget. Bounded action-quantum rounding of a proposed quarter-to-three-quarter split preserves the one-fifth property whenever the quantum is smaller than $h_\delta/20$. Otherwise precision must be increased; a fixed-precision run is not allowed to ignore this requirement.

For (iii), let $\bar{x}$ be the acquired state. Its downward displacement is at most $d2^{-b}$ in ℓ^1, and $c(\bar{x}) \le c(x)$ preserves feasibility. At a fixed action the stage cost is $8/d$-Lipschitz and the transition has column sums at most $11/16$. Thus replacing $\bar{x}$ by x changes Q_r by at most $(8 + 11\beta d G_{r-1}/16)2^{-b}$. The corresponding value change is at most $dG_r 2^{-b}$. Add these two allowances to (i). Substituting the policy's own future yields the usual one-step loss recursion; backward induction gives (16). No policy-value oracle and no population claim about fitting are used. *Q.E.D.*

2.8. *Proof of the result 4*

PROOF: A routed action is merely another feasible partition point. The endpoint and minorant induction in Theorem 7 therefore applies without change. After each evaluation retain the smallest upper endpoint and the maximum of the old and newly valid global lower endpoints. Their difference cannot increase. The stopping test uses this difference, not a fitted value or a prediction error.

Put $h_\delta = \sqrt{3\delta/H_r}$. A partition interval of length at most h_δ has minimum minorant at least its smaller lower endpoint minus $3\delta/8$. The associated upper endpoint, with the stipulated width and capacity allowance, is within δ of that minorant.

Such an interval cannot be responsible for an unresolved global gap. Every split parent therefore has length greater than h_δ. The one-fifth condition makes each of its children longer than $h_\delta/5$. Starting with one interval, disjoint terminal fragments consequently number at most $1 + 5c(x)/h_\delta$. Counting endpoints and allowing for rounding of this numerical bound gives the displayed conservative budget. Floating failure is handled by the separately charged exact-rational procedure, not by this real-arithmetic count.

The first assertion about predictor evaluation follows from the order of the stopping test and the routing rule. For the last assertion, use backward induction on the remaining horizon, with the common analytic terminal kernel as base. At each nonterminal query, declined predictions leave the same conventional split, lower bound, witness and tie rule. Induction over partition refinements gives the same next query and returned endpoints; the induction over dates then gives the same implemented policy. Model-evaluation work can still differ, so transcript identity is not a zero-cost fitting claim. *Q.E.D.*

2.9. *Proof of Proposition 5*

PROOF: Every retained lower endpoint is at least its true endpoint value minus w, and hence at least $f(a^*) - w$. On an interval of length at most h, the parabolic minorant is therefore at least $f(a^*) - w - Hh^2/8$. Subtracting the omitted-capacity allowance gives $L \geq f(a^*) - w - Hh^2/8 - \Lambda\Delta$; retaining an older, larger valid lower bound only improves this inequality. The upper endpoint at $\widehat{a}$ is at most $f(\widehat{a}) + w$. Semiconcavity and the one-sided first-order condition give $f(\widehat{a}) - f(a^*) \leq \lambda e + He^2/2$. Subtract the lower bound to prove (7). At an interior optimum the derivative is zero. The example $f(a) = a$ on $[0, c]$ shows that the boundary linear term is needed. Finally, retaining bounds makes $U' \leq U$ and $L' \geq L$; expansion proves (8). *Q.E.D.*

2.10. *Proof of the result 1*

PROOF: For each drawn bin sequence, conditional integration over its continuous realizations places the exact path expectation between the interval endpoints.

Uniform mixing over the equal bins recovers the original continuous law. The lower endpoint expectation is thus no larger than D_i, and the upper endpoint expectation is no smaller. Apply the bounded empirical-Bernstein inequality of Maurer and Pontil (2009) to the endpoint samples and take the union bound, with outward arithmetic included in the mean, variance and radius. Conditional validity for every $\mathcal{H}$ implies unconditional validity by integration. On this one simultaneous event, all ensuing deterministic comparisons of the covered quantities hold together. *Q.E.D.*

2.11. *Proof of the result 2*

PROOF: Subtract the gain intervals from k and compare the resulting interval endpoints with zero. Taking the image of $[l, u]$ under the absolute-value map gives (3). The assertions hold simultaneously for every nonnegative fee on the original coverage event. *Q.E.D.*

2.12. *Proof of the result 3*

PROOF: Subtract the common J_{b_0}. The selected implementation has net difference at most $U(\lambda)$, and the best catalogue implementation has net difference at least $L(\lambda)$. This proves the regret bound pointwise on the one simultaneous event and hence for every λ. A minimum of finitely many affine functions changes its active affine expression only at an intersection. For $\lambda_2 > \lambda_1$, sum the two minimizing inequalities to obtain $(\lambda_2 - \lambda_1)(s_{\widehat{b}(\lambda_2)} - s_{\widehat{b}(\lambda_1)}) \le 0$. Since s_b increases strictly with b, the precision comparative static follows. *Q.E.D.*

3. PROOFS OF THE OPERATIONAL AND NEW RESULTS

3.1. *Proof of the result 8*

PROOF: Strong convexity gives $f((1-s)l + su) \le (1-s)f(l) + sf(u) - \mu s(1-s)(u-l)^2/2$. Replacing the endpoint values by their upper bounds proves (18). Subtraction of a valid lower bound proves (19). A failed screen adds no endpoint to the partition and alters no lower bound. Subsequent protected refinement is therefore covered by the earlier termination proof. A successful screen supplies

exactly the local accuracy premise used in the acquired-policy recursion. Finally, C is a convex quadratic in a, proving the sublevel-set assertion. *Q.E.D.*

3.2. *Proof of the result 9*

PROOF: A successful screen returns without an endpoint query. A failed screen invokes a procedure bounded by $K(z)$, and its margin loss equals one. This proves (22). For each fixed predictor the variable $K(z)\ell_\tau(z,\theta)$ lies in $[0, K_{\max}]$. The bounded-sum inequality of Hoeffding (1963) and a union bound over the conditionally fixed family give (23). *Q.E.D.*

3.3. *Proof of the result 10*

PROOF: The chord is a convex quadratic in its normalized action, which is affine in w. Squaring the positive part preserves convexity and gives a continuously differentiable function. The nonnegative weighted sum and quadratic penalty preserve these properties. Convexity gives $F(w) - F(v) \le g^\top (w - v)$, and the minimum of $g^\top v$ on the box is $\sum_j \min\{0, g_j\}$. This proves (25). Every failed screen has squared positive-part residual at least τ^2, while every successful screen uses zero further endpoint queries. Summing the corresponding K_i bounds proves the work claim. Strong convexity with modulus λ proves the distance bound. *Q.E.D.*

3.4. *Proof of the result 11*

PROOF: Semiconcavity means that $f(a) - H\|a\|_2^2/2$ is concave. Its barycentric inequality, with vertex values replaced by lower endpoints, gives (26). The tangent inequality for the convex quadratic p proves (27); a linear function attains its simplex minimum at a vertex. Taking minima over a cover preserves the lower bound. Each feasible upper witness is an action of the actual model. The local gap premise therefore implies the earlier policy recursion, irrespective of how the triangle or support point was proposed.

For finiteness, use simplex lattice denominator A, with $(A+1)(A+2)/2$ vertices. On triangles of diameter proportional to $c(x)/A$, the curvature error tends to zero as A^{-2}. The endpoint error is reduced by finite descendant accuracy and

quadrature refinements. The retained exact-rational recovery contract requires $\delta > 16\Lambda 2^{-32}$ at each recursive call, leaving positive slack after rounding; dyadic enlargement of A and q then terminates. This is a finite arithmetic procedure, not a uniform bound on rational bit cost. For (28), the independent innovation variances after conditioning on the bins are $\rho_j^2/(3q^2)$, their direction norms squared are d, and the semiconcavity Taylor remainder is $M_{t+1}/2$ times their sum. Substitution of $\rho_1 = 1/32$, $\rho_2 = 1/64$ gives the coefficient. Finally, each action query has q_r^2 descendants, proving (29). *Q.E.D.*

3.5. *Proof of the result 1*

PROOF: Conditional expectation and $b_{t+1} = b_t\beta_t$ imply

$$\mathbb{E}\sum_{t<T} b_t a_t^{\pi}(X_t^{\pi}) = \mathbb{E}\left[\sum_{t<T} b_t c_t(X_t^{\pi}, \pi_t(X_t^{\pi})) + b_T g(X_T^{\pi}) - V_0(X_0^{\pi})\right].$$

The intermediate value terms cancel. Subtracting the two expectations cancels the initial value under the common initial law, proving the identity. Summing the pointwise local bounds proves the range. Applying the bounded-sum inequality to each lower and upper endpoint sample, with failure allocation $\alpha/(2M)$, and taking a union bound proves (2). Endpoint containment supplies the comparison between endpoint expectations and the target. The width identity follows by subtraction. No pathwise identity between D and the realized cumulative-cost difference is asserted. *Q.E.D.*

3.6. *Proof of the result 2*

PROOF: The sample-size condition implies $h_n \le s$. Insert this and the numerical-width condition into the width identity. The final claims are set inclusions for the simultaneous confidence interval. *Q.E.D.*

3.7. *Proof of the result 8*

PROOF: The bounded-sum argument in Theorem 9 gives $Pf_\theta \le u_\theta$ for every θ on one event of probability at least $1-\alpha$. For $0 \le f \le K_{\max}$, the layer-cake represen-

tation gives $|Pf - Qf| \le K_{\max}\|P - Q\|_{\mathrm{TV}}$. Hence, on that event,

$$\mathbb{E}[N_{\theta_b}(Z_b) \mid \mathcal{F}_{b-1}] \le \min\{K_{\max}, u_{\theta_b} + K_{\max}\rho_b\}.$$

The variables $c_b N_{\theta_b}(Z_b)$ have conditional ranges of length at most $c_b K_{\max}$. Conditional Hoeffding bounds, multiplied successively along the filtration and followed by the exponential Markov inequality, show that the sum exceeds its conditional means by at most the final term of (11), except on an event of probability γ. Union with the validation failure event proves the result. Every selected readout remains a proposal to the same verified gate. Its statewise certificate, and thus the earlier acquired-policy recursion, is unchanged. *Q.E.D.*

3.8. *Proof of the result 9*

PROOF: The expected cost difference in a normal block is $m\Delta$; it is zero in a declined block. Linearity of expectation gives $m(B - D)\Delta$, without requiring block independence. Multiply the cost interval by its nonnegative price, and add the work difference and fee. All price choices use the same event, so no additional probability allocation is needed. *Q.E.D.*

4. RETAINED INTERVENING TRAINING RESULTS

4.1. *Proof of Proposition 6*

PROOF: A successful screen incurs no recovery queries. On a failed screen the clipped loss equals one, and recovery work is at most WK. The pointwise bound therefore holds for the bounded variable $WK\ell_\tau$. The bounded-sum inequality and a union bound give the claim. None of the separately named service costs is included in the definition of recovery work. *Q.E.D.*

4.2. *Proof of Proposition 7*

PROOF: The lower cover still covers the entire feasible set, so its unchanged lower bound remains valid. Each additional upper endpoint bounds the true cost of its own feasible action, hence is an upper bound for the infimum. Their minimum

preserves this property. The returned action's true cost is no greater than its own upper endpoint, proving the regret claim. *Q.E.D.*

5. PROOFS FOR FROZEN CONTINUATIONS AND PURCHASED WITNESSES

5.1. *Frozen-continuation comparison*

PROOF OF THEOREM 12: Fix the construction information. Conditional expectation at date t gives

$$\mathbb{E}[b_t \beta_t P_t^{\pi_t} h_{t+1}(X_t)] = \mathbb{E}[b_{t+1} h_{t+1}(X_{t+1})].$$

Summing these equalities cancels all intermediate continuation terms in (30); the terminal correction gives

$$\mathbb{E} Z_h^\pi = \mathbb{E} J_0^\pi - \mathbb{E} h_0(X_0^\pi).$$

The common initial law cancels the last expectation when policies are compared. Their trajectory coupling otherwise does not matter for this identity.

Pointwise, $Q_t^h(x, \pi_t(x)) - h_t(x) = r_t^h(x) + a_t^{h,\pi}(x)$. For any two trajectory realizations, the difference of their residual terms is between $-\omega_t$ and ω_t. Their greedy-gap difference is between $-\eta_t^\nu$ and η_t^π. The same oscillation argument applies to $g - h_T$ at the terminal states. Multiplication by the nonnegative discounts and summation proves (32).

For the regret assertion, choose constants $\underline{r}_t, \overline{r}_t$ enclosing r_t^h with difference at most ω_t; an arbitrarily small enlargement handles unattained infima and suprema. Likewise enclose the terminal correction by $\underline{z}, \overline{z}$. For every feasible policy μ, $a_t^{h,\mu} \geq 0$. Telescoping therefore gives, at each initial state,

$$J_0^\mu \geq h_0 + \sum_{t<T} b_t \underline{r}_t + b_T \underline{z}, \qquad J_0^\pi \leq h_0 + \sum_{t<T} b_t (\overline{r}_t + \eta_t^\pi) + b_T \overline{z}.$$

Taking the infimum over μ in the first inequality, subtracting it from the second, and then removing the enlargement proves (33). No minimizing action or optimal-policy attainment is needed. If h_t is replaced by $h_t + k_t$, then $Q_t^h - h_t$ increases by $\beta_t k_{t+1} - k_t$, and the terminal correction decreases by k_T. Their discounted sum changes

by $-k_0$, identically on both trajectories. Residual oscillations and greedy gaps are unchanged. With $h = V^*$ all residual and terminal oscillations vanish. *Q.E.D.*

For completeness, the confidence statement is conditional on both policy and continuation construction. After intersection with the true support, each endpoint sample lies in an interval of length R_h. A lower-tail or upper-tail bounded-sum bound with failure allocation $\alpha/(2M)$ gives the two ends of (34). Endpoint containment implies $\mathbb{E}L_i \leq \mathbb{E}D_h \leq \mathbb{E}U_i$. A union bound proves simultaneous coverage. Dependence within a paired trajectory is permitted; independence is required across the sampled rows under this construction. A predictor or cache modified in response to earlier inference rows is not conditionally fixed by the original training information.

5.2. *Exact conditional moments and uniform qualification*

PROOF OF PROPOSITION 13: If U is uniform on $[-A, A]$, integration of $(z+u)_+$ gives

$$\frac{1}{2A}\int_{-A}^{A}(z+u)_+\,du = \frac{(z+A)_+^2 - (z-A)_+^2}{4A}.$$

A second integration with respect to an independent uniform variable on $[-B, B]$ integrates each positive-part square to a positive-part cube divided by three. Dividing by both uniform densities gives (36). Vanishing radii correspond to deterministic projections and must be removed before division. Symmetry of the innovation permits each nonzero projection coefficient to be replaced by its absolute value.

In a rational shallow network every projected center and radius is rational. Positive parts, powers, sums and divisions by checked nonzero rationals are exact rational operations. The affine part has its usual conditional mean, and linearity of integration allows signed readout coefficients. Computing all K projections at a state and action takes $O(Kd)$ arithmetic operations; each subsequent one- or two-shock neuron integral takes a fixed number of operations. Summation along n trajectories of length T gives the stated count. This says nothing about the size of the numerators and denominators or about the work used to construct and qualify h. *Q.E.D.*

Proof of the residual qualification account. Let $v_t^h(x) = \inf_{a \in A(x)} Q_t^h(x,a)$. Given an action $a \in A(y)$, proportional reduction to $A(x)$ has ℓ^1 displacement $(\sum_i a_i - c(x))_+$, at most $(c(y) - c(x))_+ \le \kappa \|x - y\|_1$. Hence

$$v_t^h(x) \le Q_t^h(y,a) + (L_{x,t} + \kappa L_{a,t}) \|x - y\|_1.$$

Taking the infimum over a, and reversing x, y, proves the asserted modulus of v_t^h. Subtracting h_t proves that of r_t^h. Every point of a radius-ρ state cover is within ρ of a cover point, so its endpoint expansion yields (37). For an action mesh with fill radius s, approximate any feasible action by a feasible mesh action within s. The Lipschitz inequality gives $\min_{\text{mesh}} Q_t^h - L_{a,t} s \le v_t^h \le \min_{\text{mesh}} Q_t^h$. Outward action endpoints preserve both inequalities. There are $N_t A_t$ such conditional evaluations per date. Actor-specific greedy gaps and the terminal correction are additional certificates, not consequences of checking only the residual at the cover points.

5.3. *Transport and actual-policy validity*

PROOF OF THEOREM 17: The value modulus gives $V_r^*(x) \ge L_y - L_V D$. Nonnegative economic costs give the zero lower bound. Variation first in state and then in action yields

$$Q_r^*(x, \widetilde{a}) \le Q_r^*(y, a_y) + L_Q D + L_A \|\widetilde{a} - a_y\|_1 \le U(x).$$

Both state endpoints and the repaired action lie in the ambient state–action domain on which these moduli are valid. Feasibility of $\widetilde{a}$ supplies $V_r^*(x) \le Q_r^*(x, \widetilde{a})$. Outward arithmetic preserves the two end inequalities; testing the resulting width, rather than an inward-rounded width, gives the local gap.

At each history and date, the returned action has original-optimum advantage no greater than its certified local gap and the stated acquisition allowance. Conditional expectation of the next value therefore gives the same one-step regret recursion used in Theorem 7. Backward induction proves the full discounted loss sum, even when the action selected at x depends on which valid records earlier requests purchased. No equality with the fresh-search action is used. In particular, a population policy-

cost interval for a different, frozen controller cannot be borrowed merely because both controllers have the same local-gap upper bound. *Q.E.D.*

The moduli can also be checked directly from the inherited primitive proof. The state cost has ℓ^1 Lipschitz constant $8/d$; the transition's induced ℓ^1 state norm is at most $11/16$. The action cost on the ambient nonnegative simplex of total mass at most $1/4$ has component derivatives bounded by $13/16$, and each exposure column has ℓ^1 norm $3d/8$. Composition with the G_{r-1}-Lipschitz true continuation gives L_Q, L_A. The value modulus G_r is Theorem 4. These constants do not depend on a fitted predictor.

5.4. *Capacity rounding and the fill bound*

PROOF OF PROPOSITION 18: Let $c_q(x) = q\lfloor c(x)/q \rfloor$. If an already lattice-aligned a_y has total mass no greater than $c_q(x)$, it is unchanged. Otherwise proportional reduction removes total mass $\sum_i (a_y)_i - c_q(x)$, and downward rounding removes less than mq more. Since $c_q(x) > c(x) - q$ and $a_y \in A(y)$, the total displacement is at most $\kappa D + (m+1)q$.

Before the favorable zero truncation of the lower bound, the expanded width is no greater than

$$\gamma\delta + (L_V + L_Q + \kappa L_A)D + L_A(m+1)q + \xi.$$

It is at most δ for $D \le \rho_*$. Because all records are checked and retained, two centers at which fresh fills are purchased must consequently be more than ρ_* apart in ℓ^1. Partition each coordinate of the unit cube into $n = \lceil d/\rho_* \rceil$ equal intervals, with a fixed boundary convention. The diameter of each resulting box is at most $d/n \le \rho_*$. It can contain at most one fresh center. Thus $M_r \le n^d \le (1 + d/\rho_*)^d$. Each fill is paid once, whereas every query incurs its actual lookup, verification and record handling. Adding these operations proves (45). If records are evicted or invalidated, the separation argument applies only within a period in which the required earlier records remain available; lost coverage must be repurchased and charged. *Q.E.D.*

6. PROOFS FOR QUALIFICATION-OPTIMAL READOUTS AND SEQUENTIAL COSTS

6.1. *The readout program and its dual*

PROOF OF THEOREM 14: The readout has Lipschitz constant at most $L(\theta) = \sum_k L_k|\theta_k|$. At a cover point the residual is in $[y_j^- - F_j\theta, y_j^+ - F_j\theta]$. The Lipschitz constant of that residual is at most $C + L(\theta)$. For any state choose a cover point at distance at most ρ. Its lower endpoint minus $\rho(C + L(\theta))$ is a valid lower bound there, and the upper endpoint plus that quantity is a valid upper bound. Taking the minimum and maximum over the cover gives the asserted oscillation bound. No distribution of states, interpolation of a discontinuous actor, or convexity of the fitted continuation is used.

Introduce $u, l \in \mathbb{R}$ and $s_k \geq 0$ and minimize

$$u - l + 2\rho C + \sum_k \lambda_k s_k$$

subject to

$$u \geq y_j^+ - F_j\theta, \qquad l \leq y_j^- - F_j\theta \qquad (j = 1, \ldots, N),$$

$$s_k \geq \theta_k, \qquad s_k \geq -\theta_k \qquad (k = 1, \ldots, K),$$

and $-B_k \leq \theta_k \leq B_k$. The stated dimensions follow. For any θ the tight choices of u, l and the choice $s_k = |\theta_k|$ attain (38); conversely every feasible point has objective at least that expression. Continuity on the compact coefficient box gives attainment.

The maximum of finitely many real numbers is the maximum of their averages over probability vectors. Consequently the objective can be written as

$$2\rho C + \max_{p,q \in \Delta_N} \left\{ p^\top y^+ - q^\top y^- + [F^\top (q - p)]^\top \theta + \sum_k \lambda_k |\theta_k| \right\}.$$

For a scalar z,

$$\min_{|v| \leq B} \{zv + \lambda|v|\} = -B(|z| - \lambda)_+.$$

Indeed zero is minimizing when $|z| \leq \lambda$, whereas the appropriate endpoint of the coefficient interval is minimizing otherwise. Applying this identity coordinatewise

proves weak duality in (39). The finite program is feasible and has a finite attained optimum. Linear-programming duality, or equivalently minimax on the compact coefficient box and the two probability simplexes, interchanges the minimum and maximum in the last display. It gives equality for some p, q. For any explicitly supplied feasible triple, subtracting its dual lower bound from its primal objective bounds its excess over the optimum. *Q.E.D.*

Rational verification and the backward qualification chain. The empirical implementation uses rational feature values, rational label endpoints and rational moduli. A numerical optimizer proposes coefficients and dual multipliers. Coefficients are rounded to a fixed dyadic lattice and clipped to the declared box. Nonnegative dual proposals are rounded and normalized to sum exactly to one; a fixed unit vector is the valid fallback if their mass vanishes. The objective, support function and optimization gap are then recomputed in rational arithmetic. Solver failure or a gap exceeding the declared tolerance is recorded rather than converted into an assertion of optimal fitting.

At date t this argument conditions on the actual h_{t+1} produced by the next date's computation. Repeating backwards gives valid bounds $\omega_t \le \mathcal{W}_t$ simultaneously because they are deterministic qualifications, not confidence statements. Substituting them into the frozen-continuation proof gives

$$\Omega_h \le \sum_{t<T} b_t \mathcal{W}_t + b_T \omega_T, \qquad J_0^\pi - V_0^* \le \sum_{t<T} b_t \mathcal{W}_t + b_T \omega_T + H_h^\pi.$$

The same substitution is valid in the cost-contrast support. This does not compare the future continuations generated by different backward optimization choices, and hence makes no joint optimality assertion.

6.2. *Whole-cell policy qualification in the executed construction*

For the new two-state construction, the original terminal payoff is retained exactly. Each subsequent continuation is additive with nine coefficients: a constant and four features per state coordinate. The ReLU family uses x_j and $(x_j - k/4)_+$, $k = 1, 2, 3$. The conventional family uses the four nonconstant degree-four Bernstein terms

$\binom{4}{k} x_j^k (1-x_j)^{4-k}$, $k = 1, \dots, 4$. Feature Lipschitz bounds are respectively one and four. The latter follows by differentiating a Bernstein term: its derivative is four times the difference of two nonnegative degree-three Bernstein terms whose sum over the full basis is one. Both families have the same coefficient box and node information; their spanning spaces and modulus bounds need not coincide.

The midpoint cover with n cells per coordinate has radius $\rho = 1/n$ in ℓ^1. If the actual future has Lipschitz constant L_+, the inherited primitive inequalities give

$$L_x = 4 + \beta \frac{11}{16} L_+, \qquad L_a = \frac{13}{16} + \beta \frac{3}{4} L_+, \qquad C = L_x + \frac{L_a}{16}, \quad \beta = \frac{15}{16}.$$

At the terminal continuation $L_+ = 5$; at an earlier date it is the verified coefficient-weighted feature modulus. The 33-point action mesh at a midpoint x has exact fill radius $c(x)/64$. Thus subtracting $L_a c(x)/64$ from its minimum action value gives the lower continuous-action label, while the minimum itself is the upper label. All those evaluations are rational conditional expectations under the original uniform innovation law.

For the implemented actor, a second 33-point menu is rounded down to the $1/4096$ lattice and bounded by the minimum capacity over the whole observation cell. Let a_j minimize its exact midpoint action value and let y_j^- be the continuous-action lower label. At any state x in that cell,

$$\begin{aligned} Q^h(x, a_j) - \inf_a Q^h(x, a) &\le Q^h(x_j, a_j) - y_j^- + (L_x + C)\|x - x_j\|_1 \\ &\le Q^h(x_j, a_j) - y_j^- + (L_x + C)\rho. \end{aligned}$$

Taking the maximum over all cells gives the recorded uniform greedy-gap allowance. Replacing a_j by zero gives the comparator allowance. The selected action is feasible throughout its cell, including its closed boundary, and a fixed boundary convention gives a well-defined actor. This proves the continuous-state, continuous-action qualification; midpoint evaluations alone would not do so.

For a ReLU future, exact conditional integration follows Proposition 13. The independent checker instead splits the innovation interval at every hinge crossing and integrates each affine piece by its endpoint trapezoid. For a Bernstein future, the production calculation expands the uniform moments. The independent checker

uses the positive five-point weights $(7,32,12,32,7)/90$ on equally spaced points from $-1/32$ to $1/32$. Substitution of each monomial of degree zero through five verifies this rule directly, so it integrates the conditional degree-four polynomial exactly. The checker separately uses the retained original rational terminal primitive at every final-date action value. These checks concern all new construction nodes and actions, not the entire historical recursive Bellman computation.

6.3. *A time-uniform endpoint bound*

PROOF OF THEOREM 15: Condition on $\mathcal{F}_{i-1}$. The frozen-continuation telescoping proof applies to the two complete controllers selected for this episode; it gives $\mathbb{E}[D_i \mid \mathcal{F}_{i-1}] = \Delta_i$. History-dependent actions cause no change in the tower property provided the actual conditional transition kernels and all required greedy-gap qualifications remain valid. The common initial law cancels the initial continuation term. The terminal correction is still required.

Write $R_i = b_i - a_i$. Since $L_i \le D_i \le U_i$ and the endpoints lie in $[a_i, b_i]$,

$$\mathbb{E}[L_i \mid \mathcal{F}_{i-1}] \le \Delta_i \le \mathbb{E}[U_i \mid \mathcal{F}_{i-1}].$$

Conditional Hoeffding's lemma therefore gives, for each fixed $\lambda > 0$,

$$\mathbb{E}\left[\exp\left\{\lambda(L_i - \Delta_i) - \frac{\lambda^2 R_i^2}{8}\right\}\middle| \mathcal{F}_{i-1}\right] \le 1.$$

To see the constant directly, a variable supported on an interval of length R has tilted variance at most $R^2/4$. The second derivative of its centered log moment-generating function is this tilted variance. Integrating twice from zero bounds that log moment-generating function by $\lambda^2 R^2/8$. The nonpositive conditional mean of $L_i - \Delta_i$ can only decrease the exponential bound. The same reasoning applies to $\Delta_i - U_i$.

Products of these conditional factors are nonnegative supermartingales starting at one. Ville's inequality bounds the probability that the lower-endpoint product for λ_j ever exceeds $2/(\alpha w_j)$ by $\alpha w_j/2$. The same bound holds for the upper-endpoint product. A union bound over both signs and all j gives simultaneous validity with

failure probability at most $\alpha \sum_j w_j \le \alpha$. Outside that exceptional event every partial sum is bounded by every corresponding linear boundary. Taking their minimum gives (40) and (41). The predictable support also bounds each Δ_i, justifying the additional intersection.

The event already covers every finite index, so evaluating it at a finite stopping time does not consume additional error probability. With a fixed complete controller pair and fixed initial-law/reset contract, its conditional expected original-law contrast is the same number Δ at every episode. Predictable changes of a valid critic alter the observations and ranges, not this identity. Without that fixed target, the preceding calculation identifies only the displayed conditional average. *Q.E.D.*

The implementation need not evaluate exponentials. For $x \ge 1$, write $x = 2^k y$ with $1 \le y < 2$, and use

$$\log y = 2 \sum_{r=0}^{m-1} \frac{z^{2r+1}}{2r+1} + \mathcal{R}_m, \qquad 0 \le \mathcal{R}_m \le \frac{2z^{2m+1}}{(2m+1)(1-z^2)}, \quad z = \frac{y-1}{y+1}.$$

Use the same expression at $z = 1/3$ for $\log 2$. It gives an exact rational upper bound on each logarithm in the boundary; replacing the logarithm by that bound enlarges the interval. Arithmetic operation counts do not bound rational bit complexity.

The finite set of bets is deliberate: a finite minimum is directly computable. It remains time-uniform for arbitrarily many episodes, although a finite bet set can leave a positive asymptotic radius. Increasing the predeclared set improves its supported scales but is a new design choice, not a free after-the-fact selection. One may instead preallocate a countable summable family; an implementation using any finite subset retains validity with its assigned weights.

6.4. *Stopping, failed episodes and the replacement decision*

PROOF OF COROLLARY 16: On the simultaneous event, $l_N \le \Delta \le u_N$. Adding the declared incremental charge and the one-time charge divided by the deployment count gives the claimed interval. Its upper endpoint being negative implies strict net cost reduction. Inclusion in an equivalence band is a different two-sided assertion.

The event supports every such deterministic algebraic implication, including a stopping rule that inspects the interval repeatedly.

For a new epoch, condition on the history at its start, which includes the construction and selection of its new controller pair. Apply the theorem with that epoch's previsible error allocation α_e. A conditional union bound followed by expectation bounds the probability of any erroneous qualification by $\mathbb{E}\sum_e \alpha_e$, no larger than the total declared budget. No independence between epochs is required. *Q.E.D.*

A support reserved before an episode remains valid when its numerical evaluation fails. Recording the full interval $[a_i, b_i]$ then preserves endpoint containment and the martingale argument. Deleting the episode after an outcome-dependent failure need not preserve the conditional law. All attempted work is retained irrespective of which interval is recorded. An invalid support certificate is different: it supplies no admissible substitute observation and must not be treated as a harmless numerical timeout.

Finally, a conditional kernel-bias bound can be accommodated only when it is itself justified: if $|\mathbb{E}[D_i \mid \mathcal{F}_{i-1}] - \Delta_i| \le \zeta_i$ with previsible certified ζ_i, enlarge both ends by $n^{-1}\sum_{i\le n}\zeta_i$. This follows by applying the proved bound to the actual conditional mean and then using the triangle inequality. An empirical model discrepancy without such a bound is not set to zero.

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